AI-AUGMENTED METHODS FOR INTERPRETABLE AND EFFICIENT MODELING IN MODERN ASTRONOMY

25-26J-108

Project Proposal Report

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INTERPRETABLE GALAXY FORMATION MODELING VIA PHYSICS-INFORMED NEURAL NETWORKS (PINNS) AND EXPLAINABLE AI (XAI)

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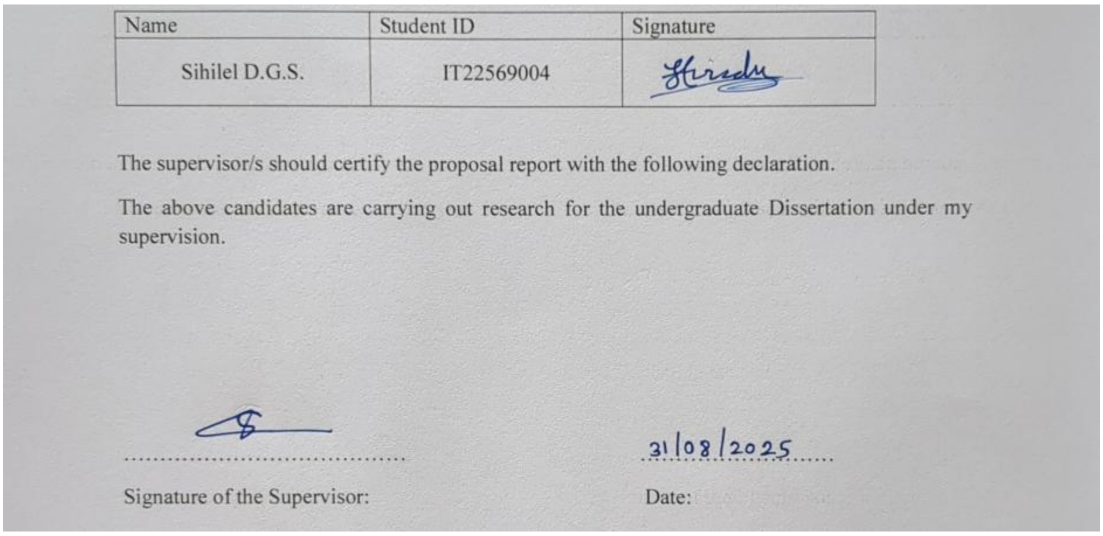
DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.



Name	Student ID	Signature
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The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the Supervisor:

Date: 31/08/2025

DEDICATION

This work is dedicated to our supervisor, Mr. Samadhi Chathuranga; our co-supervisor, Dr. Kapila Dissanayake; our external supervisor, Mr. Banuka Dimuthu; and to all the friends and family whose guidance, encouragement, and support have been invaluable throughout this journey.

ACKNOWLEDGEMENT

I wish to express my sincere gratitude to my supervisor Mr. Samadhi Chathuranga Rathnayake for invaluable guidance, constructive feedback, and encouragement throughout the research. I also thank my co-supervisor Dr. Kapila Dissanayake, and all my peers, for their assistance and insightful discussions.

ABSTRACT

Understanding the birth and evolution process of galaxies is among the primary pursuits in modern astronomy. Vast surveys such as the Sloan Digital Sky Survey (SDSS) and the Mapping Nearby Galaxies at Apache Point Observatory (MaNGA) program have generated terabytes of observable data, affording unprecedented opportunity to study galaxy evolution. Most modern machine learning models, though, treat these datasets as "black boxes" — excelling in making accurate predictions while giving very little insight into whether or not their predictions conform to the known astrophysical laws.

The current research introduces an Interpretable Galaxy Formation Modeling framework in the form of **Physics-Informed Neural Networks (PINNs)** with **Explainable Artificial Intelligence (XAI)** approaches. The model will exclusively be trained on actual observational datasets and will involve astrophysical regulations as constraints in training. By integrating these physical regulations in the model's learnings, we will increase trustworthiness and maintain predictions in agreement with known galaxy formation principles.

The PINN will be compared to traditional classical black-box models in terms of predictive accuracy as well as interpretability. XAI tools such as SHAP values, concept activation, as well as counterfactual reasoning will provide explanations for model selections in terms that are astronomer friendly.

Keywords: galaxy formation, physics-informed neural networks, explainable AI, astronomy

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LIST OF ABBREVIATIONS

Abbreviation	**Meaning**
AI	Artificial Intelligence
ALFALFA	Arecibo Legacy Fast ALFA Survey
APO	Apache Point Observatory
CSV	Comma-Separated Values
ESA	European Space Agency
FITS	Flexible Image Transport System
GAMA	Galaxy And Mass Assembly Survey
GRINN	Gravity-Informed Neural Network
HPC	High Performance Computing
HST	Hubble Space Telescope
JWST	James Webb Space Telescope
LIME	Local Interpretable Model-agnostic Explanations
LSST	Legacy Survey of Space and Time (Vera C. Rubin Observatory)
MaNGA	Mapping Nearby Galaxies at Apache Point Observatory
ML	Machine Learning
NGC	New General Catalogue (of Nebulae and Clusters of Stars)
PINN	Physics-Informed Neural Network
PI-AstroDeconv	Physics-Informed Astronomical Image Deconvolution
SDSS	Sloan Digital Sky Survey
SHAP	SHapley Additive exPlanations
SPINN	Simulation with Physics-Informed Neural Network
TCAV	Testing with Concept Activation Vectors
XAI	Explainable Artificial Intelligence
ZTF	Zwicky Transient Facility

ASCII	American Standard Code for Information Interchange
xGASS	The extended GALEX Arecibo SDSS Survey

1. INTRODUCTION

1.1 BACKGROUND AND LITERATURE SURVEY

The birth and evolution of galaxies is a foundational challenge in astrophysics, connecting the early universe physics to the variety in structure that we see today. There is an extraordinary variety in the morphology, luminosity, and stellar populations in galaxies, from huge ellipticals that are old-stellar-dominated to actively forming stars, gas-rich spirals to systems that are irregular due to tidal encounters. All this variety is an outcome of the accumulation in cosmic time of gravitational collapse, retention of angular momentum, hydrodynamical effects, star birth, chemical enrichment, as well as merging.

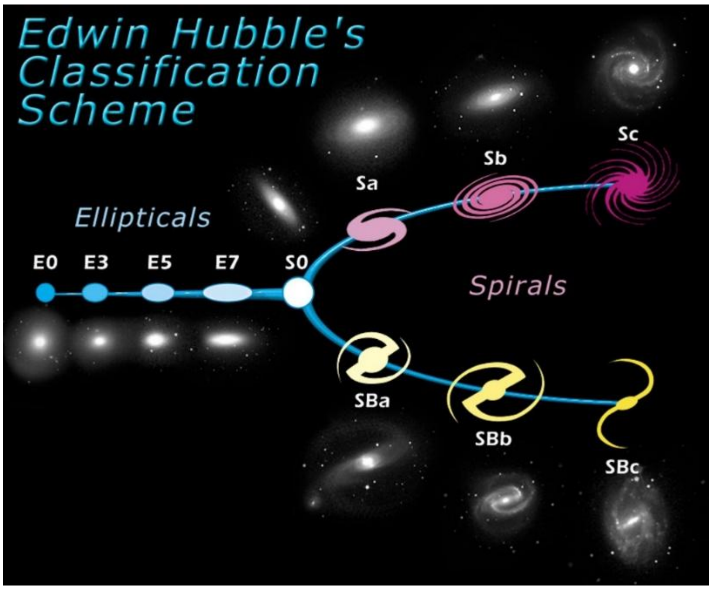

Figure 1: Diagram of Hubble's Tuning Fork classification scheme. This is a classical type of image still used by astronomers today to show how galaxies are classified. Credit: Space Telescope Science Institute

Following the Big Bang, matter throughout the universe was distributed nearly evenly, having small variation in density. These fluctuations formed under the influence of gravity, collapsing matter into virialized shapes. The resultant morphologies depended strongly on the angular momentum of the proto-galactic clouds: high angular momentum created rotationally supported disks, while low angular momentum created pressure-supported spheroids. Star birth, typically compelled through mergers or accretion events, spanned varied luminosities and colors, enriched heavy elements in the interstellar medium, as well as influenced succeeding stellar generations.

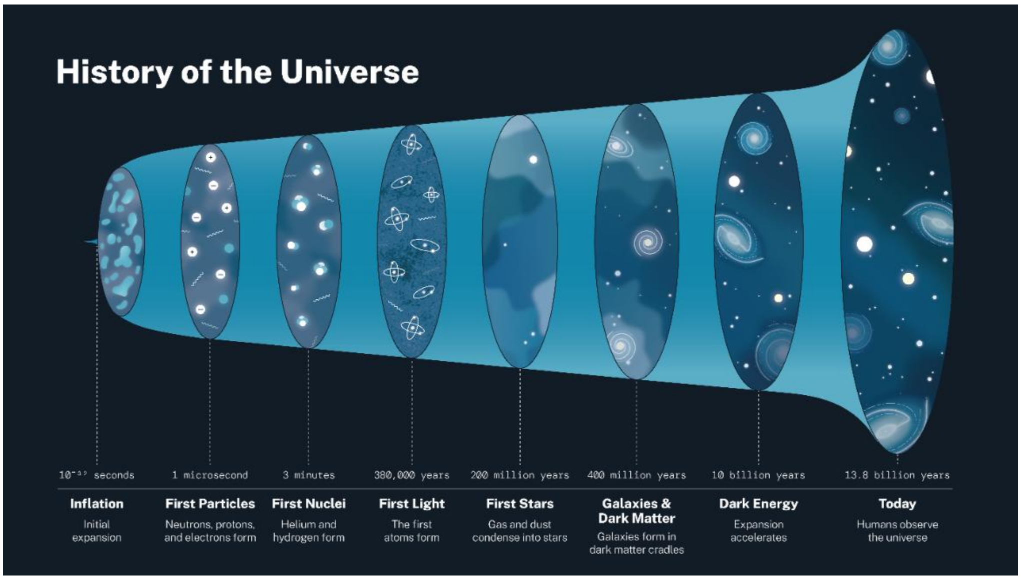

Figure 2: History of the universe. Credit: National Aeronautics and Space Administration (NASA)

Observational signatures from local as well as high-redshift surveys have provided significant evolutionary patterns. Big galaxies like ellipticals and spirals already existed by $z \approx 1$, whereas small irregular galaxies experienced rapid variations in luminosity and number abundance in the past 8 billion years. Populations like Lyman-break galaxies ($z \gtrsim 2.5$) are progenitors that give rise to current spheroids, but exact connections between late-stage and early-stage systems are uncertain as a consequence of gaps in the observational coverage as well as intricacy in underlying processes.

Figure 3: The Mice or NGC 4676, they will eventually merge into a single giant galaxy. Credit: NASA, H. Ford (JHU), G. Illingworth (UCSC/LO), M. Clampin (STScI), G. Hartig (STScI), the ACS Science Team, and ESA

The primary task is to create models that capture the pertinent physics of galaxy evolution while being tractable and falsifiable. A number of cosmological simulations rely on complex sub-grid recipes for feedback and star formation that, while numerically successful, can obscure the physical origins of measurable quantities. To avoid this, this research employs Physics-Informed Neural Networks (PINNs), which incorporate governing gravity, hydrodynamical, and star-forming equations into the learning. Fitted with explainable model techniques in the form of feature attribution as well as saliency mapping, this technique attempts to discover how the initial conditions, motions in the gas, as well as merger history individually dictate galaxy morphology, luminosity history, as well as history of star formation.

It will incorporate physically motivated constraints in combination with interpretability to yield more transparent theory-observation linkages that will provide understanding into mechanisms that dictate galaxy evolution over cosmic time.

1.1.1 Use of Physics-Informed AI in Astronomy

Physics-Informed Neural Networks (PINNs) and related physics-guided AI methods merge the pattern-recognition ability of machine learning with established physical laws. Instead of relying on correlations in data alone, these models are constrained by rules such as conservation of mass, momentum, or energy, ensuring their predictions remain physically realistic. These approaches have shown strong results in other fields—including climate science, engineering, and fluid dynamics—and are now being explored in astronomy for tasks ranging from galaxy modeling to cosmological simulations and data interpretation.

In **cosmology and galaxy modeling**, PINNs are being used to replicate the outputs of computationally expensive simulations at a fraction of the cost, while retaining physical accuracy. For instance, Dai *et al.* (2023) developed a PINN that "filled in" baryonic (normal matter) properties in dark matter–only simulations by embedding empirical relations—such as the link between galaxy mass and its halo—into the model [1]. This resulted in more accurate predictions of stellar mass, gas content, and metallicity than purely data-driven baselines. Similarly, Korber *et al.* (2023) introduced PINION, which modeled the evolution of hydrogen ionization in the early universe by directly enforcing the governing chemical equations [2]. With minimal training data, PINION reproduced large-scale reionization patterns seen in detailed radiative transfer simulations.

In **complex dynamical systems**, PINNs are being applied to problems like "fuzzy" dark matter, whose behavior resembles waves rather than particles. Mishra & Tolley (2025) created SPINN, a solver that embeds both gravity and quantum pressure equations, capturing both large-scale structure and fine interference patterns [3]. Auddy *et al.* (2023) applied a similar method to simulate self-gravitating gas flows, finding their GRINN model matched traditional codes while scaling more efficiently in higher dimensions [4].

In **orbital mechanics**, Saz Ulibarrena *et al.* (2023) combined Hamiltonian Neural Networks—designed to conserve energy—with standard numerical solvers [5]. This hybrid approach used the neural network for speed, but fell back to the physics-based solver if energy conservation was violated, offering a stable, faster solution for large systems such as asteroid swarm simulations.

PINNs have also shown promise for **observational data analysis**. Ojha *et al.* (2024) developed LensPINN, which incorporates the gravitational lens equation directly into a deep learning model, enabling consistent reconstructions of both background galaxies and lens mass distributions [6]. Ni *et al.* (2024) created PI-AstroDeconv, embedding a telescope's point spread function into the network to improve image deblurring without needing ground-truth data, while recovering key scientific measurements like the hydrogen power spectrum more reliably [7].

Taken together, these studies illustrate how physics-informed AI can unite the adaptability of machine learning with the rigor of physical theory. By embedding constraints such as gravitational laws, fluid dynamics, or optical system responses, researchers are building models that are faster, more accurate, and more interpretable than black-box alternatives. **However, despite these advances, large-scale applications to galaxy formation remain rare—particularly those that integrate physical consistency, interpretability, and direct use of real observational data—leaving a clear gap for further research.**

1.1.2 Interpretable AI in Astronomy

Interpretable AI methods—such as SHapley Additive exPlanations (SHAP), Local Interpretable Model-agnostic Explanations (LIME), and concept-based analysis—are designed to make machine learning models more transparent. They help scientists understand which inputs most influence a model's predictions and whether those inputs align with known science. In astronomy, these tools have been used for tasks like classifying galaxy shapes [8] and estimating photometric redshifts [9], allowing researchers to check that the model's reasoning matches physical expectations.

For instance, SHAP has been applied to measure how much features such as galaxy color, brightness, and structure contribute to classification decisions—showing, for example, that some color indices are far more predictive for early-type galaxies than for late-type ones. LIME has been used in redshift estimation to identify which wavelength bands most affect the output, highlighting regions where the model may be less reliable. Concept-based tools like Testing with Concept Activation Vectors (TCAV) have linked model activations to broader astrophysical properties, such as star formation rate or dust content, providing a way to connect learned features to established concepts.

These methods have improved trust in AI systems for astronomy, but they are almost always applied separately from physical modeling. Most work focuses on either explaining black-box models without enforcing physical realism, or on building physics-informed models without offering detailed explanations of how predictions are made. Bringing these two goals together—by creating AI systems that are both interpretable and grounded in physical laws—would enable models that are not only accurate, but also transparent and scientifically credible. This combination is especially important for complex problems like galaxy formation, where understanding *why* the model predicts a certain outcome is as valuable as the prediction itself.

1.2 RESEARCH GAP

Lack of physical constraints in existing models

Many machine learning (ML) models used in astronomy focus on maximizing predictive accuracy but do not ensure their outputs obey established physical laws. These models often behave as “black boxes,” finding patterns in data without checking if the results are physically plausible according to known rules of how galaxies form and evolve. For example, purely data-driven deep learning models in other domains have been shown to produce physically inconsistent outputs (e.g., predicting negative values for inherently positive quantities) when no physical constraints are imposed, underscoring the need to integrate scientific laws into the modeling process [10].

Studies in various fields have demonstrated that injecting physical knowledge or constraints into ML models can improve both accuracy and realism of predictions – improving performance while preventing unphysical results [10]. In astrophysics, the benefits of physics-informed ML are just beginning to be explored. Even advanced galaxy modeling frameworks such as UniverseMachine [11], which calibrates semi-empirical galaxy formation prescriptions to match many observed properties, can still produce outcomes that agree with some observations but conflict with others. For instance, UniverseMachine had to assume an unusually high fraction of “orphan” galaxies (satellites with no dark matter halo) in order to reproduce certain galaxy clustering data – an assumption that is at odds with standard physical expectations for galaxy evolution [11] [12]. Such discrepancies highlight the need for new ML methods that make predictions while also guaranteeing physical consistency with known laws of nature.

Limited ability to explain predictions

Another critical gap is the difficulty of interpreting predictions from modern ML models. Traditional models such as decision trees allowed researchers to identify which galaxy properties (e.g., mass, color, morphology) drove predictions. By contrast, deep neural networks like convolutional networks obscure their decision processes behind millions of parameters, making it challenging to determine whether the model relies on scientifically meaningful features [12]. Recent work, such as Wu et al. (2024) [13], introduced Sparse Feature Networks to produce more interpretable features. However, most ML approaches to galaxy formation still do not combine interpretability tools with physics-based constraints, limiting their reliability and scientific transparency.

Heavy reliance on simulated data

Many high-profile ML studies in galaxy research are trained mainly on data from computer simulations rather than real telescope observations. Simulations can produce large, labeled datasets, but they are always approximations of reality. For example, a deep learning model trained on simulated galaxies was able to classify real Hubble Space Telescope images, but the simulations it learned from did not include some important processes like black hole feedback, meaning the model could be biased. Models trained only on simulations may also pick up on simulation-specific quirks that don't exist in the real universe. At the same time, running detailed simulations is expensive, which limits how much training data can be produced. Large observational surveys such as the Sloan Digital Sky Survey (SDSS) already provide millions of real galaxy images, but very few ML frameworks train and validate only on real data. This leaves a gap for approaches that learn directly from real observations, making them more immediately applicable to real-world astronomy.

In summary, current ML work in this area rarely delivers all three of the following at once:

- **Physical consistency** – ensuring predictions follow established physical laws.
- **Interpretability** – allowing scientists to see why a model made a prediction.
- **Grounding in real data** – training and validating primarily on observational datasets.

Some recent efforts have started combining real observational data with explainable AI methods like SHAP values to connect model features to physical galaxy properties. However, these are exceptions rather than the rule. There is still a clear opportunity to build models that are both physically informed and interpretable, and that work directly with large real-world datasets.

Research Gap Summary

Researc h	Main Focus	Physical Constrain ts	Interpretabili ty	Real Observation al Data	Simulate d Data Reliance	Addresse s All Three Gaps?
Lahav (2023)	Review of AI in cosmology	✗	✓	✓	✗	✗
Lieu (2025)	Comprehensi ve review of interpretable AI in astronomy	✗	✓	✓	✗	✗
Wu et al. (2024)	Sparse Feature Networks for interpretable galaxy analysis	✗	✓	✓	✗	✗
Lockha rt et al. (2020)	PINN in planetary space physics	✓	✓	✓	✗	✗
Stephen s (2018)	Deep learning on simulated galaxies to classify real HST images	✗	✓	✓	✓	✗

Table 1: Research gap comparison

1.3 RESEARCH PROBLEM

Current AI applications in galaxy formation modeling face a fundamental trust problem. Even when these models produce accurate predictions, scientists cannot be sure that:

1. The outputs obey known physical laws,
2. The reasoning behind predictions is transparent, and
3. The results are valid for real galaxies rather than only for simulated ones.

Most existing work either focuses on prediction accuracy without physical safeguards [11] [12], or applies interpretability tools to classification and parameter estimation tasks without embedding physical constraints [9]. In the few cases where physics-informed methods are used in cosmology [2] [3], they are rarely applied to galaxy formation, and almost never in combination with interpretability techniques and real observational training data.

This creates a clear and urgent need for a modeling framework that:

- **Uses observational data** (e.g., from SDSS, HST, JWST) as the primary source for learning and validation,
- **Incorporates physics-informed constraints** specific to galaxy formation processes such as gas inflows, mergers, and feedback, and
- **Provides transparent explanations** for predictions, enabling scientists to see how model inputs lead to specific outputs.

Without such a framework, our ability to connect theory, simulation, and observation in galaxy formation research will remain limited. The problem is not just achieving high predictive accuracy but achieving it in a way that is physically consistent, interpretable, and applicable to real galaxies.

As such, the research question for this project would be: **Can physics-informed neural networks, combined with explainability tools, offer interpretable and physically consistent simulations of galaxy formation?**

2. RESEARCH OBJECTIVES

2.1 MAIN OBJECTIVES

The overarching goal of this research is to create a next-generation galaxy formation modeling framework that is accurate, physically consistent, and fully interpretable. While existing machine learning models can achieve high predictive performance, they often lack safeguards to ensure their outputs follow established scientific principles, and they rarely provide clear explanations for their predictions. This project addresses these shortcomings by combining Physics-Informed Neural Networks (PINNs) with Explainable AI (XAI) techniques, trained directly on real observational datasets from major astronomical surveys.

Specifically, the main objectives are:

1) **Develop a physics-informed model for galaxy formation prediction**
   Build a neural network framework that embeds astrophysical laws directly into its learning process, ensuring predictions remain consistent with known relationships while adapting flexibly to observational data.
2) **Incorporate scientific rules as soft constraints during training**
   Represent key astrophysical relations—such as mass–luminosity scaling and gas fraction trends—as penalty terms or regularization functions in the loss function. This allows the model to balance accuracy with physical plausibility, rather than sacrificing one for the other.
3) **Integrate explainability into the prediction process**
   Use XAI techniques to trace predictions back to their most influential input features, enabling astronomers to verify whether the model is focusing on meaningful physical drivers or on spurious patterns. This ensures that the model not only predicts well but also provides scientifically interpretable reasoning.
4) **Benchmark against conventional "black-box" models**
   Compare the proposed framework with state-of-the-art purely data-driven models on both predictive performance and interpretability. The evaluation will quantify trade-offs between accuracy, physical realism, and explanation quality, providing a clear measure of the added value from physics-informed, interpretable modeling.

2.2 SPECIFIC SUB OBJECTIVES

1) **Acquire and preprocess galaxy formation data from major astronomical surveys**

Gather datasets from observational programs such as MaNGA, Arecibo Legacy Fast ALFA Survey (ALFALFA), The extended GALEX Arecibo SDSS Survey (xGASS), and Galaxy And Mass Assembly Survey (GAMA). This involves cleaning, standardizing, and integrating multi-source data, ensuring compatibility across different formats and measurement systems. The prepared dataset will serve as the foundation for training and validating the PINN.

2) **Embed astrophysical scaling laws into the learning framework**

Reformulate key galaxy formation relations—such as mass–luminosity scaling, virial equilibrium, and gas fraction correlations—into explicit constraint terms for the model. These constraints will be guiding the network toward physically consistent predictions while reducing overfitting to purely data-driven trends.

3) **Integrate explainable AI (XAI) techniques to interpret model predictions**

Apply model interpretability tools, including SHAP values, counterfactual analysis, and concept activation methods, to identify the most influential input parameters driving galaxy property predictions. The insights gained will be compared with established astrophysical understanding, enabling traceable connections between input features and predicted outcomes.

4) **Evaluate the PINN against predictive accuracy, physical realism, and interpretability metrics**

Assess model performance using three complementary evaluation dimensions:

- **Predictive metrics** (e.g., RMSE, R^2) for accuracy on test data.
- **Physical-consistency metrics** that quantify adherence to embedded astrophysical laws.
- **Interpretability metrics** measuring the clarity, stability, and fidelity of explanations generated by XAI methods.

These evaluations will be benchmarked against conventional black-box models to determine trade-offs between accuracy and interpretability.

3. METHODOLOGY

This study applies a physics-informed machine learning framework to model aspects of galaxy formation while maintaining interpretability through explainable AI techniques. The methodology integrates astrophysical theory, computational modeling, and modern data science practices to ensure scientific validity and transparency in the results.

3.1 REQUIREMENTS GATHERING AND ANALYSIS

The research began with a review of scholarly literature on galaxy evolution, with emphasis on physical mechanisms such as gravitational collapse, hydrodynamic processes, star formation histories, and merging events. Sources included peer-reviewed astrophysics journals, data archives, and published simulation datasets. Input from domain experts in astronomy was sought through online discussions to refine the project scope and identify feasible physical constraints for integration into the neural network model. These requirements informed the selection of variables, data sources, and computational approaches for the study.

3.2 FEASIBILITY STUDY

3.2.1 Schedule Feasibility

The research is structured to be completed within the one-year time frame specified for the final-year project. The timeline allocates distinct phases for data acquisition, preprocessing, model development, training, explainability analysis, and evaluation. Each milestone is aligned with interim deliverables to ensure measurable progress.

3.2.2 Technical Feasibility

The proposed approach requires combining physics-informed neural networks (PINNs) with explainable AI methods to simulate and interpret galaxy formation processes. Implementation will use high-level programming languages such as Python, alongside numerical and astrophysics-oriented libraries (e.g., NumPy, SciPy, TensorFlow/PyTorch, Astropy). Available computational resources are sufficient for training on moderate-scale datasets, with provisions to use preprocessed subsets for efficiency.

The methodology will embed astrophysical laws—such as mass conservation, virial equilibrium, and simplified gravitational dynamics—into the model's loss functions. This ensures outputs are physically consistent while reducing overfitting to purely data-driven correlations. Explainability will be incorporated using techniques like saliency maps, SHAP values, and gradient-based attribution, enabling analysis of which physical parameters most influence simulated galaxy properties.

3.3 TOOLS AND TECHNOLOGIES

This study will leverage following tools, software frameworks and libraries.

- Python
- Pandas
- NumPy
- Astropy
- PyTorch
- SHAP
- Captum
- Jupyter Notebook

Figure 4: Tools used in the research

3.4 SYSTEM DIAGRAM

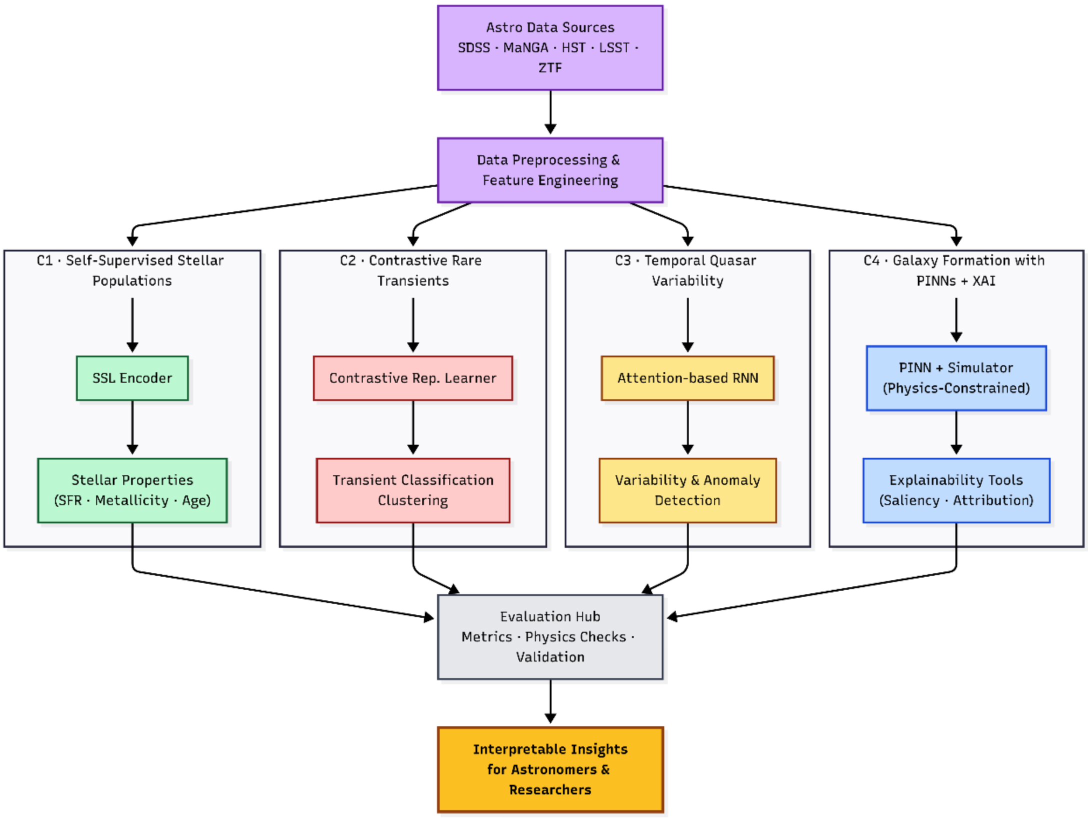

Figure 5: System Architecture Diagram

The system architecture illustrates how the four independent research components in the project are organized, while still sharing a common foundation. At the top, astrophysical data sources (e.g., SDSS, MaNGA, HST, LSST, ZTF, etc) are collected and standardized through a data curation and feature store. From there, the data flows into four parallel research pipelines, each representing a distinct mini-project:

- **C1 – Self-Supervised Stellar Populations:** learns latent representations from galaxy spectra and photometry to infer stellar properties such as star formation rates, metallicity, and age.
- **C2 – Contrastive Learning for Rare Transients:** applies contrastive methods on light curves to improve classification and clustering of uncommon astronomical transients.
- **C3 – Temporal Quasar Variability:** uses attention-augmented recurrent models to capture quasar luminosity variability patterns and anomalies linked to black hole activity.

- **C4 – Galaxy Formation via PINNs + XAI (your component):** integrates physical constraints into a neural network simulator and explains results with XAI methods (saliency maps, attributions).

The outputs from all pipelines are fed into a shared evaluation hub, where models are benchmarked against physical consistency checks and observational metrics. Results then pass through an interpretability and visualization layer, ensuring that outcomes are transparent and scientifically valid.

Finally, all four pipelines converge into a common outcome layer, producing interpretable insights for astronomers and researchers. This structure emphasizes both the independence of each research strand and their alignment toward a shared goal of physically consistent, explainable astrophysical modeling.

3.5 COMPONENT DIAGRAM

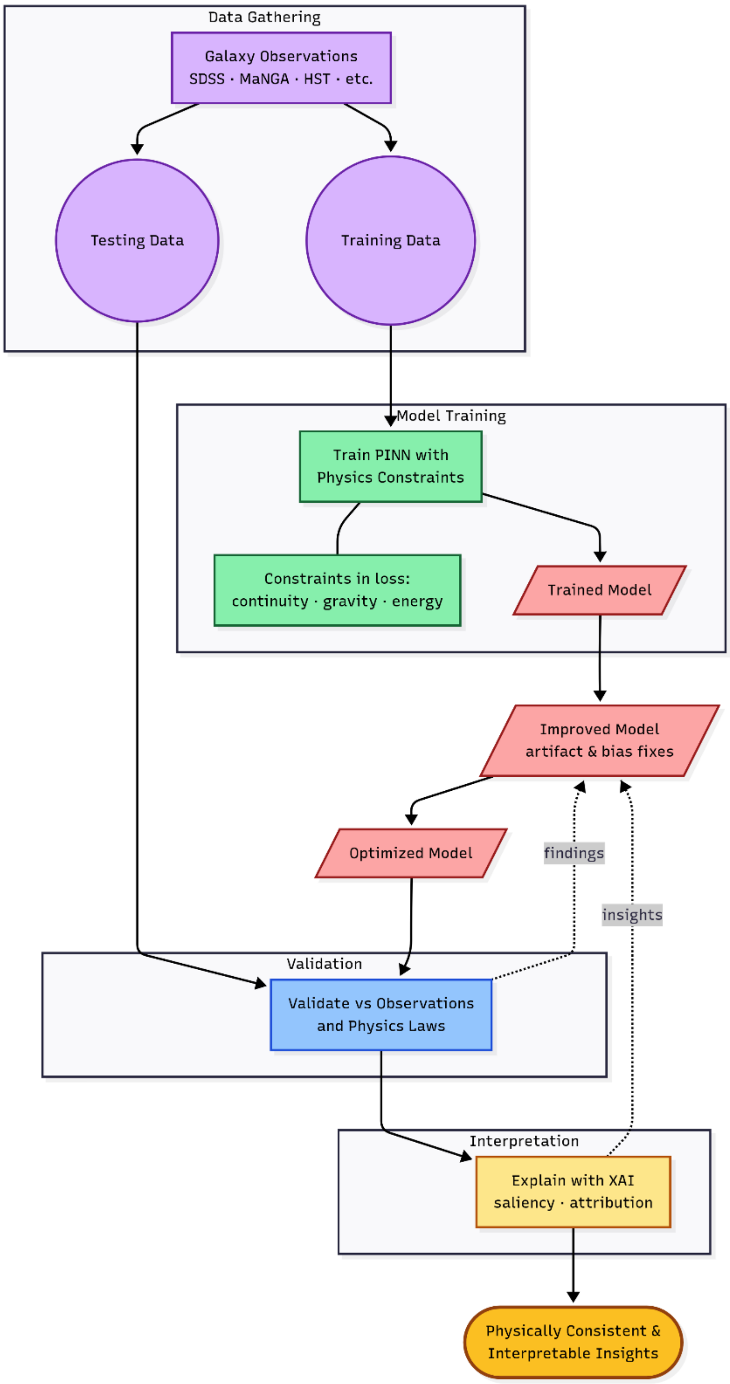

Figure 6: Component Diagram

The process begins with galaxy observations (e.g., SDSS, MaNGA, HST), which are divided into training and testing datasets. The training data is used to develop a Physics-Informed Neural Network (PINN), where physical laws such as gravity, hydrodynamics, and energy conservation are embedded directly into the model's loss functions.

The initial trained model is refined through iterative improvement loops, addressing observational artifacts, data biases, and optimization strategies. The final optimized model is then evaluated against the testing data, with performance measured both in terms of predictive accuracy and adherence to physical consistency checks.

Following validation, the results are passed through an explainability module, where tools such as saliency maps, gradient attributions, and perturbation analyses provide insights into how the model makes its predictions. The final outcome of this component is a physically consistent and interpretable model of galaxy formation, offering astronomers both reliable predictions and transparent reasoning.

3.6 PROJECT REQUIREMENTS

Functional Requirements

These define what the system must do — the features and operations needed to achieve the research objectives.

1. **Data Acquisition and Management**
    - Ability to collect galaxy property data from multiple astronomical surveys (e.g., MaNGA, ALFALFA, xGASS, GAMA).
    - Support for importing data in different formats [Flexible Image Transport System (FITS), Comma Separated Values (CSV), American Standard Code for Information Interchange (ASCII) tables].
    - Tools for cleaning, standardizing, and integrating datasets into a unified format.
2. **Physics-Informed Model Development**
    - Implement a Physics-Informed Neural Network (PINN) architecture.
    - Encode astrophysical laws (e.g., mass–luminosity relation, virial theorem) as soft constraints in the model's loss function.
    - Support both supervised learning (predicting galaxy properties) and physics-based regularization.
3. **Explainable AI Integration**
    - Incorporate SHAP, LIME, and TCAV for feature attribution and model interpretation.
    - Generate visual explanations (e.g., feature importance plots, counterfactual examples).
    - Provide a report mapping influential features to known astrophysical concepts.
4. **Evaluation and Benchmarking**

- Compare PINN model performance against baseline "black-box" models (e.g., standard deep neural networks).
- Use predictive accuracy metrics (RMSE, R^2), physical consistency scores, and interpretability measures.
- Store evaluation results in a structured format for analysis and reporting.

5. **Result Visualization and Reporting**
   - Produce plots comparing predicted vs. observed galaxy properties.
   - Summarize adherence to embedded physical laws.
   - Generate an interpretable, publishable output report.

Non-Functional Requirements

These define quality attributes, constraints, and performance criteria.

1. **Performance**
   - Training time should be optimized to handle large observational datasets within reasonable compute limits.
   - The system should scale to datasets containing hundreds of thousands of galaxies without significant slowdowns.
2. **Usability**
   - Code should be documented for ease of reuse by other researchers.
   - Output visualizations and explanations must be understandable by astronomers without ML expertise.
3. **Reliability**
   - The model should produce consistent predictions when retrained on the same dataset.

- Physics constraints should be enforced in all training runs to prevent unphysical results.

4. **Maintainability**
    - The system should be modular, allowing easy updates to physical constraints, training datasets, or interpretability methods.
5. **Portability**
    - The framework should be deployable on different computing environments (local machine, HPC cluster, or cloud).
6. **Reproducibility**
    - All experiments must be accompanied by fixed random seeds, documented parameters, and dataset versioning.

3.7 VALIDATION

Validation is essential to ensure that the developed Physics-Informed Neural Network (PINN) produces galaxy formation predictions that are both scientifically credible and aligned with real-world observations. In this project, validation will be carried out by directly comparing model outputs with high-quality observational data from large astronomical surveys such as MaNGA, ALFALFA, xGASS, and GAMA.

This process will evaluate three aspects:

1. **Predictive Accuracy** – Assessing how closely the predicted galaxy properties match observed values using statistical metrics (e.g., RMSE, R^2).
2. **Physical Consistency** – Ensuring predictions adhere to embedded astrophysical laws (e.g., mass–luminosity relations, gas fraction trends) and identifying any violations.
3. **Interpretability Alignment** – Using Explainable AI methods to confirm that the features driving predictions correspond to established astrophysical understanding rather than spurious correlations.

Discrepancies between predictions and observations will be carefully analyzed. Where mismatches occur, constraint formulations or network parameters will be refined to improve both physical plausibility and predictive reliability. This iterative validation–refinement cycle will help build confidence in the model's outputs, making it a trustworthy tool for understanding galaxy formation.

3.8 WORK BREAKDOWN STRUCTURE

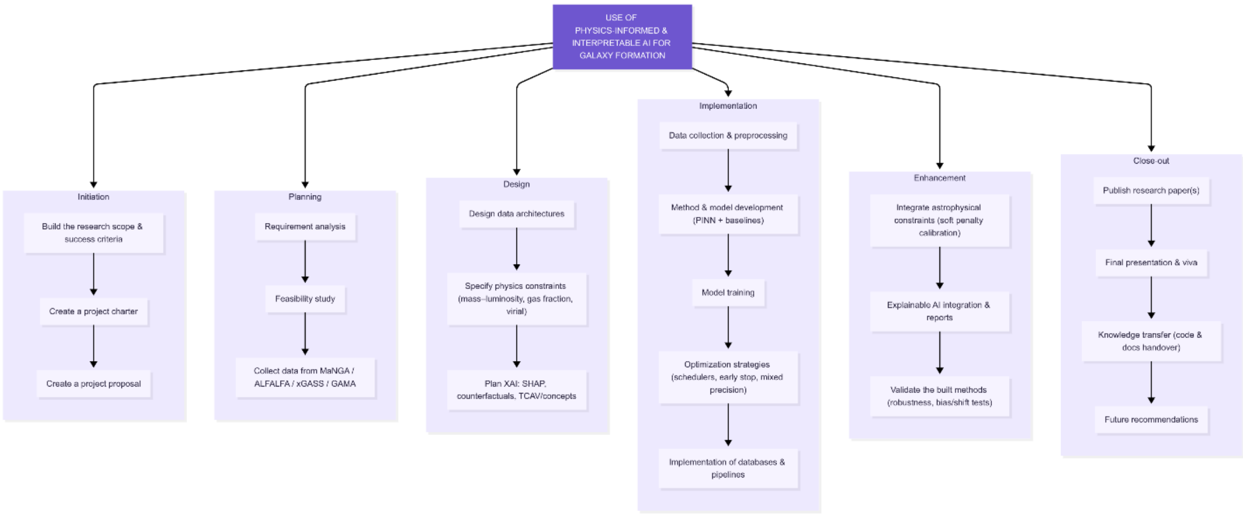

Figure 7: Work Breakdown Structure

The Work Breakdown Structure (WBS) organizes the Interpretable Galaxy Formation via PINNs + XAI project into six major phases: Initiation, Planning, Design, Implementation, Enhancement, and Close Out. Each phase is further broken into specific tasks.

- Initiation covers defining the research problem, creating the project charter, and drafting the proposal.
- Planning includes requirement analysis, feasibility studies, and preparing the data collection and preprocessing plan.
- Design focuses on constructing the PINN architecture, formulating physics-informed loss functions, and planning the integration of XAI methods.
- Implementation involves data ingestion, model training, optimization, and experiment tracking.
- Enhancement is dedicated to iterative improvements such as artifact mitigation, bias correction, integration of XAI tools, and validation.
- Close Out finalizes the work with publications, presentations, documentation, and recommendations for future research.

3.9 GANTT CHART

The Gantt chart presents the project timeline from August 2025 to June 2026, showing key phases such as proposal development, model training, validation, XAI integration, and reporting. Each milestone, including progress presentations and final deliverables, is mapped sequentially to ensure clear tracking of research activities and deadlines.

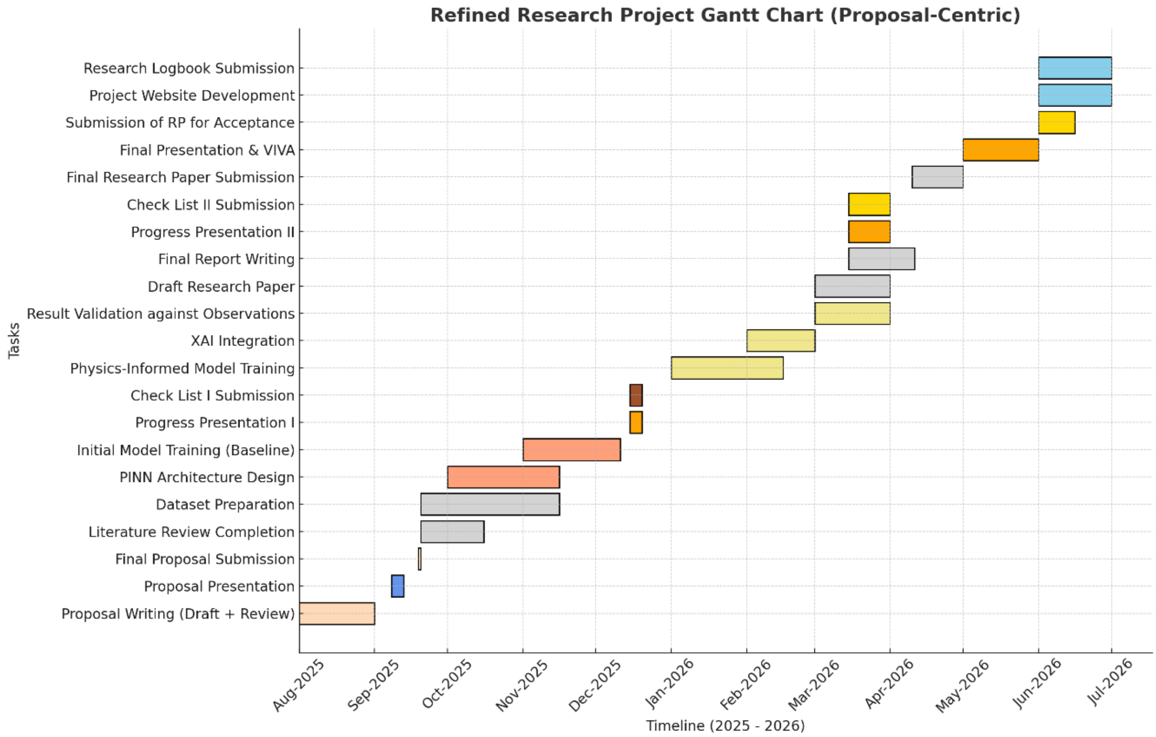

Figure 8: Project Gantt Chart

4. COMMERCIALIZATION AND CONTRIBUTION TO THE DOMAIN

The techniques developed in this project—combining Physics-Informed Neural Networks (PINNs) with Explainable AI (XAI) for galaxy formation modeling—offer direct value to organizations engaged in space science, exploration, and long-term resource utilization. Agencies such as NASA, European Space Agency (ESA).

By embedding physical constraints such as gravity, hydrodynamics, and dark matter interactions into machine learning models, this framework provides physically reliable simulations that remain consistent with established astrophysical laws. At the same time, the integration of XAI ensures that predictions are interpretable, allowing researchers and engineers to trace model outputs back to the underlying drivers. This dual emphasis on realism and transparency reduces the risks associated with "black-box" predictions and strengthens confidence in using AI to support high-stakes decision-making.

Practical applications include:

- Data interpretation: Interpretable outputs enable astrophysicists to link observational data to underlying formation processes, improving accuracy in identifying stellar populations, gas flows, or dark matter distributions.
- Resource usage for models: As an undergraduate project, these models will be optimized for freely available or community driven resources. As such, researchers can save money spent on heavy computation resources.
- Observation strategy optimization: By highlighting features that are difficult to detect directly, the models can guide more targeted observation campaigns, saving costs and increasing the likelihood of discovery.

5. DESCRIPTION OF PERSONNEL & FACILITIES

This research is supervised by Mr. Samadhi Chathuranga. He is a lecturer in the Department of Computer Science of the Faculty of Computing, Sri Lanka Institute of Information Technology (SLIIT), Malabe, Sri Lanka.

Dr. Kapila Dissanayake is a senior lecturer (higher grade) currently attached to the Department of Computer Science of the Faculty of Computing, Sri Lanka Institute of Information Technology (SLIIT), Malabe, Sri Lanka.

Mr. Banuka Dimuthu joins this research as the external supervisor, a lecturer, and a research team member at the Institute of Astronomy, Sri Lanka. Also an astronomical observer at the City Observatory, Institute of Astronomy, Sri Lanka.

This research will be conducted by the following four team members:

- Sihilel D.G.S. — will be working on Interpretable Galaxy Formation Modeling via Physics-Informed Neural Networks (PINNs) and Explainable AI (XAI).
- Senevirathne B.A.N.K. — will be working on Contrastive Learning for Classification of Rare Astronomical Transients in Multi-Band Sky Surveys.
- Dilmani H.V.N — will be working on Temporal Modeling of Quasar Luminosity Variability Using Attention-Augmented RNNs
- Gunasinghe W.M.S.H. — will be working on Self-Supervised Learning for Characterizing Stellar Populations in Remote Galaxies.

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